

Multiple integrals – 4

Line and surface integrals: Green, Stokes, Gauss

13 сентября 2026 года

Theoretical minimum.

Theorem 1 (Green). For a piecewise smooth positively oriented boundary

$$\oint_{\partial D} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

In particular $\text{vol}_2(D) = \frac{1}{2} \oint_{\partial D} (x dy - y dx)$: an area is computed by a contour integral, which is how planimeters and most olympiad area problems work.

Theorem 2 (Stokes and Gauss–Ostrogradsky). $\oint_{\partial S} \langle F, d\ell \rangle = \iint_S \langle \text{curl } F, d\vec{S} \rangle$ and $\iint_{\partial V} \langle F, d\vec{S} \rangle = \iiint_V \text{div } F dV$. Both are the same statement $\int_{\partial M} \omega = \int_M d\omega$ written in dimensions two and three, and both turn a statement about a boundary into a statement about a density.

Theorem 3 (closed versus exact). A C^1 field with $\partial_j F_i = \partial_i F_j$ has a potential on every simply connected domain, and the line integral then depends only on the endpoints. On a domain with a hole this fails: the field $\frac{(-y, x)}{x^2 + y^2}$ is closed and its integral over the unit circle equals 2π — the winding number, which counts the obstruction.

Fact 1 (flux of the Newtonian field). $\text{div } \frac{x}{\|x\|^3} = 0$ away from the origin, while the flux through any closed surface enclosing the origin equals 4π . This single example explains simultaneously the role of the singularity, the meaning of the divergence theorem and the formula for the potential of a point charge.

1. Prove Green’s formula for a region bounded by a graph in each direction, deduce the formula for the area through a contour integral, and compute

by it the area enclosed by the astroid and by one arch of the cycloid together with the axis.

2. (Putnam 1986) Suppose $f_1(x), \dots, f_n(x)$ are functions of n real variables $x = (x_1, \dots, x_n)$ with continuous second-order partial derivatives everywhere on $\mathbb{R}^n$, and suppose there are constants c_{ij} with $\frac{\partial f_i}{\partial x_j} - \frac{\partial f_j}{\partial x_i} = c_{ij}$ for all i, j . Prove that there is a function g on $\mathbb{R}^n$ such that $f_i + \partial g / \partial x_i$ is linear for all i .
3. (Putnam 1987) Let $G(x, y) = (\frac{-y}{x^2+4y^2}, \frac{x}{x^2+4y^2}, 0)$. Prove or disprove that there is a vector-valued function $F(x, y, z) = (M, N, P)$ with continuous partial derivatives for all $(x, y, z) \neq (0, 0, 0)$, with $\text{curl } F = 0$ there, and with $F(x, y, 0) = G(x, y)$.
4. Prove the divergence theorem for a region bounded by a smooth surface, compute the flux of $\frac{x}{\|x\|^3}$ through a closed surface enclosing the origin and through one not enclosing it, and deduce Gauss' law.
5. (Высшая лига 2014) Recall that the divergence of a vector field $v = (v_1, v_2, v_3)$ in $\mathbb{R}^3$ is $\text{div } v = \partial_1 v_1 + \partial_2 v_2 + \partial_3 v_3$. Let v be a smooth vector field and f a smooth function on $\mathbb{R}^3$. Prove that at every point where $v \neq 0$ there is a coordinate system in a neighbourhood U of that point in which $\text{div } v = f$ throughout U .